PROGRESS REPORT · ECSP6007

STOCK MARKET ANALYSIS & PREDICTION

AI-Based Decision Support Platform for the Indian Stock Market

Roll No.	Name of Student
B-20	Rishikesh Chikhale
B-26	Saie Amdhare
B-52	Amogh Shukla
A-56	Abhishek Malhotra

Under the guidance of Dr. Jayu Kalambe

B.Tech Electronics and Computer Science · Semester VII · 2026-27

Shri Ramdeobaba College of Engineering and Management, Nagpur

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01 / REVIEW MAP

CONTENTS

1. Introduction and motivation
2. Problem statement
3. Objectives and scope
4. Literature review and research gap
5. System architecture and dataset
6. Exploratory data analysis

THE PLATFORM

27

technical indicators feed the comparison of Gradient Boosting, ARIMA and LSTM models.

INTRODUCTION AND MOTIVATION

Market growth · tool gap · research opportunity

275%

Rise in Indian retail-investor accounts: about 4 crore (2019) to over 15 crore (2024).



A fast-growing market

More retail investors need interpretable signals.



A visible tool gap

Free platforms (NSE, Moneycontrol, Groww, TradingView) provide raw data and charts, but no prediction or reasoning.



AI can scale pattern finding

Machine learning finds patterns in large historical datasets faster and more consistently than a human analyst.



The academic opportunity

AI stock-prediction research rarely becomes practical, user-friendly systems. This project pairs model evaluation with an explainable end-to-end decision-support platform for the Indian market.

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USEFUL MARKET DATA STILL LEAVES THE DECISION TO THE USER

#	Problem	What goes wrong
1	Information overload	News, social media, earnings and analyst calls conflict; no single actionable signal.
2	No predictive intelligence	Free tools show data and charts only, with no ML-based trend or recommendation.
3	Black-box AI	Paid tools output BUY/SELL with no reasoning, so users cannot trust them.
4	Manual technical analysis	RSI, MACD and Bollinger Bands by hand are slow, error-prone and difficult to scale.
5	No model comparison	Investors rely on one method, with no easy way to compare approaches.

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04 / RESEARCH DESIGN

OBJECTIVES AND SCOPE

Build broadly · evaluate honestly · explain every signal

1. Collect and clean 7+ years of NSE OHLCV data.
2. Compute 27 technical indicators and perform EDA.
3. Train Gradient Boosting, ARIMA and LSTM models.
4. Add BUY / HOLD / SELL with confidence and SHAP explanations.
5. Deploy an interactive Streamlit dashboard.

BOUNDARIES

SCOPE

- End-of-day NSE OHLCV; live refresh every 5 minutes during market hours, 9:15 AM–3:30 PM IST.
- TVS Motor, Solar Industries and HDFC Bank.
- Short-term positional trading, 1 to 30 days; not intraday.
- Technical analysis only. News sentiment and fundamentals are future work.
- Hosting on Streamlit Community Cloud.

LITERATURE REVIEW

From ensemble accuracy to explainable finance

Author(s)	Year	Key finding	Relevance
Patel et al.	2015	Random Forest leads Indian NSE stocks (80%+).	Validates Indian ensembles.
Khaidem et al.	2016	RF + indicators: 80%+ accuracy (NYSE).	Validates features.
Selvin et al.	2017	LSTM superior on NSE data.	LSTM for India.
Lundberg and Lee	2017	SHAP: game-theory feature attribution.	Basis of XAI layer.
Fischer and Krauss	2018	LSTM beats ARIMA on S&P 500 direction.	Three-model comparison.
Bussmann et al.	2020	SHAP explanations build trust in AI finance.	Transparency-first design.

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06 / THE GAP

MODELS ARE NOT ENOUGH.

A usable platform must connect evidence, comparison and explanation.

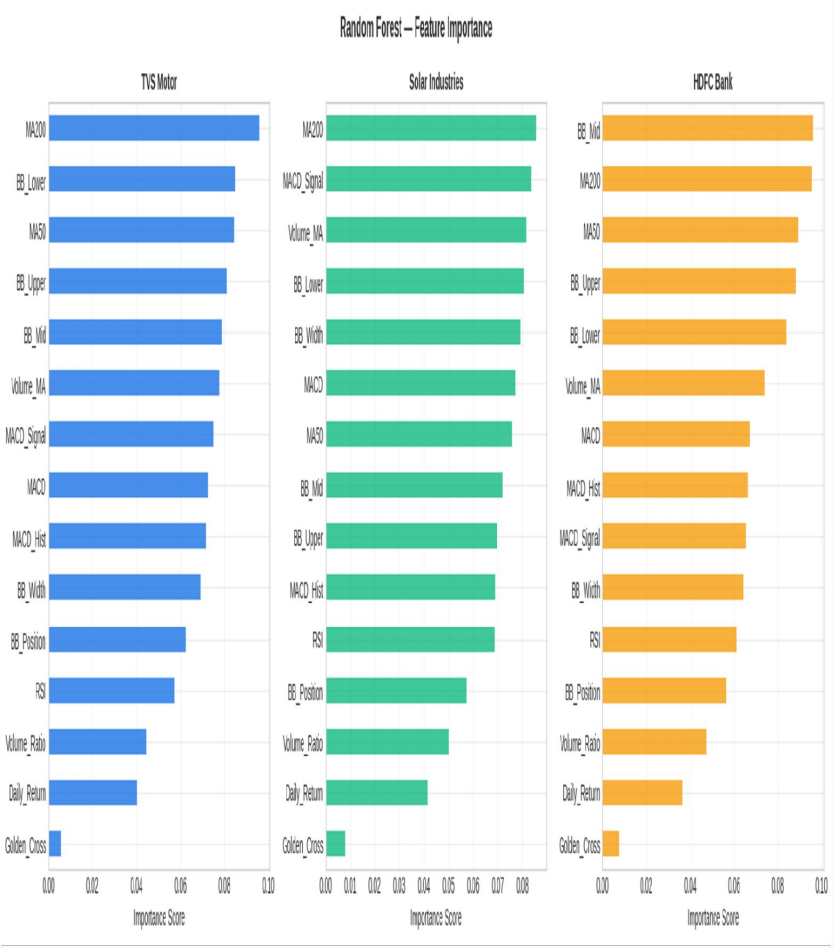
RESEARCH GAP

What existing studies leave unresolved

1. **Deployment:** all reviewed studies stop at model training and evaluation; none delivers a deployable dashboard for a non-technical retail investor.
2. **Comparison:** most studies use a single model or stock, so cross-model and cross-sector comparison is not possible.
3. **Trust:** existing AI tools are black boxes, with no explanation of their signals.

This project: Gradient Boosting, ARIMA and LSTM compared across three sectors on one live, explainable, deployable platform.

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07 / SYSTEM DESIGN

SYSTEM ARCHITECTURE

Seven layers from market data to an explainable dashboard

1. **L1 Data acquisition:** yfinance; TVSMOTOR.NS, SOLARINDS.NS, HDFCBANK.NS.
2. **L2–L3 Preparation:** cleaning, 27 indicators and UP/DOWN labels.
3. **L4 ML:** Gradient Boosting, ARIMA(3,1,1), LSTM and Linear Regression.
4. **L5 Evaluation:** Accuracy, F1, AUC, RMSE, MAPE, AIC and backtest.
5. **L6 Explainability:** SHAP TreeExplainer.
6. **L7 Presentation:** Streamlit + Plotly dashboard.

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DATASET DESCRIPTION

Yahoo Finance via yfinance · 2 Jan 2019 to 1 Oct 2026

Stock	NSE ticker	Sector	Rows	Avg close	Max close	Min close
TVS Motor	TVSMOTOR.NS	Auto and EV	1,919	Rs. 1,514.81	Rs. 4,466.20	Rs. 245.87
Solar Industries	SOLARINDS.NS	Defence	1,918	Rs. 6,000.66	Rs. 22,455.00	Rs. 837.29
HDFC Bank	HDFCBANK.NS	Banking	1,919	Rs. 713.61	Rs. 996.42	Rs. 357.66

5,756 rows

Master dataset with zero missing values; auto_adjust=True handles splits, bonus and dividends.

Seven-plus years

Includes the COVID crash, V-shaped recovery, 2021–22 bull run and 2023–24 correction.

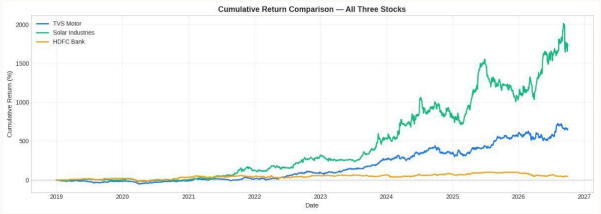
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PRICE AND RETURNS REVEAL THREE DIFFERENT MARKET BEHAVIOURS

Closing price trend · cumulative return comparison · 2019–2026



Closing price trend, 2019–2026



Cumulative return comparison

~2,000% · Solar Industries

Flat until 2023, then a sharp rise (defence spending push).

~650% · TVS Motor

Steady compounding from two-wheeler demand and EV adoption.

~50% · HDFC Bank

Drawdown in 2023–24 after the HDFC merger.

Common shock

March 2020 crash is the minimum price for all three stocks.

EDA sets the modelling context: sector, volatility and market regime matter before any prediction is made.

EXPLORATORY DATA ANALYSIS: VOLATILITY AND CORRELATION

Daily returns reveal risk; closing prices reveal co-movement.



Daily return distribution



Closing price correlation

Stock	Total return	Daily volatility	Max gain	Max loss
TVS Motor	~650%	2.005%	+14.04%	-12.76%
Solar Industries	~2,000%	2.128%	+11.62%	-11.77%
HDFC Bank	~50%	1.571%	+11.60%	-12.61%

HDFC Bank is the most stable; Solar Industries the most volatile. TVS and Solar closing prices correlate at 0.97; HDFC is 0.72 to 0.75.

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02 · FEATURE ENGINEERING

27

technical indicators

From raw OHLCV data to signals a model can learn.

Category	Features · purpose
Trend (5)	MA50, MA200, Golden Cross, Price_vs_MA50, Price_vs_MA200 — Long and short-term trend direction
Momentum (7)	RSI(14), RSI Overbought/Oversold, MACD, Signal, Histogram, MACD Positive — Speed and strength of price moves
Volatility (7)	Bollinger Upper/Mid/Lower, BB Width, BB Position, Volatility 5d and 10d — Volatility regime and price extremes
Volume (3)	Volume MA(20), Volume Ratio, Volume Spike (ratio > 1.5) — Conviction behind moves
Return (5)	Daily Return, Return 1d, 3d, 5d, 10d — Momentum across time horizons

Price and volume: Five source columns become the model input. **Indicators:** Trend, momentum, volatility, volume and return features. **Signal:** Features support UP/DOWN classification and forecast models.

RSI = 100 - 100/(1+RS); MACD = EMA(12) - EMA(26); Bollinger = 20-day SMA +/- 2 SD. The first 200 rows were dropped for MA200; RSI showed 15-20 in the March 2020 crash.

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03 · PRIMARY CLASSIFIER

MODEL 1: GRADIENT BOOSTING CLASSIFIER

Sequential ensemble learning turns feature corrections into a directional signal.

Step	Ensemble learning
Start	Train the first tree on the engineered indicators.
Correct	Each tree corrects the errors of the previous one.
Combine	300 trees create the final classifier.

Learning rate 0.05; max_depth=5; subsample=0.8; class_weight="balanced"; random_state=42. Each tree adds 5% of its correction.

TARGET AND SPLIT

Chronological, leakage-free

UP (1) if the close 5 days later is higher; otherwise DOWN (0).

70% train · 15% validation · 15% test

Signal generation

UP probability	Signal
> 55%	BUY
45% to 55%	HOLD
< 45%	SELL

Binary labels are near-balanced: TVS Motor 44.5% DOWN / 55.5% UP; Solar Industries 47.1% / 52.9%; HDFC Bank 47.6% / 52.4%.

MODELS 2, 3 AND 4: ARIMA, LSTM, LINEAR REGRESSION

Different horizons require different assumptions about price behaviour.

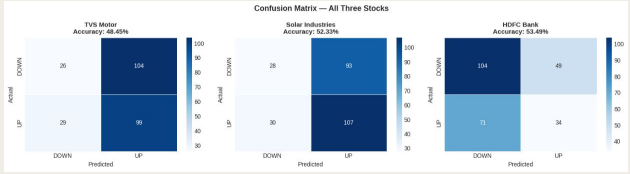
Model	Task / horizon	Settings and evidence
ARIMA(3,1,1)	7-day forecast	Fitted on the last 200 days in the dashboard; AIC: 15,706 TVS; 21,584 Solar; 12,727 HDFC; reverts to mean beyond 7 days.
LSTM	Next-day price	60-day closing-price sequences, MinMax scaled; 2 LSTM layers, 50 units; dropout 0.2; Dense 25, Dense 1; Adam lr 0.001, batch 32, early stopping patience 5; stopped at 10 / 11 / 50 epochs.
Linear Regression	30-day trend	Fitted on the last 90 trading days; confidence band widens from 1 to 3.5 volatility units; added because ARIMA is too flat for 30 days.

RESULTS: GRADIENT BOOSTING EVALUATION

The classifier finds a modest signal, with recall behaving differently across sectors.

Stock	Accuracy	Precision (UP)	Recall (UP)	F1	AUC
TVS Motor	48.45%	0.49	0.77	0.60	0.567
Solar Industries	52.33%	0.54	0.78	0.64	0.510
HDFC Bank	53.49%	0.41	0.32	0.36	0.519

Patel et al. (80%+) used a calmer pre-2015 period; recent post-2020 studies report about 52-62%.



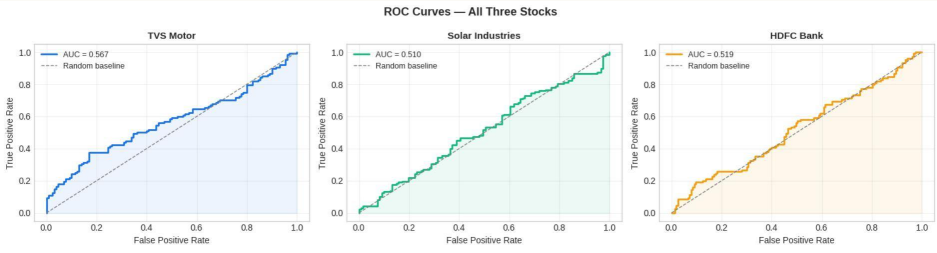
Confusion matrices: TVS, Solar, HDFC

±5% Results vary between runs due to subsampling.

- Accuracy is 48-53% against a 50% random baseline; stock direction is hard to predict in an efficient market.
- TVS and Solar models have a buying bias: high UP recall (about 77%) but low DOWN recall. HDFC Bank is the most balanced; it detects 68% of DOWN days, useful for risk management.

06 · MODEL COMPARISON

RESULTS: ROC CURVES, LSTM AND ARIMA



ROC curves with AUC scores

AUC > 0.5

A small but genuine learned signal; TVS is strongest.

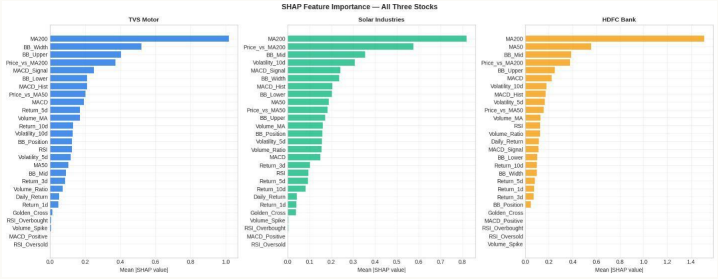
LSTM REGRESSION

Stock	AUC	LSTM RMSE	LSTM MAPE
TVS Motor	0.567	Rs. 129.14	2.57%
Solar Ind.	0.510	Rs. 734.37	3.43%
HDFC Bank	0.519	Rs. 14.43	1.33%

- LSTM MAPE: below 5% on all three stocks; HDFC Bank lowest, 1.33%.
- Solar RMSE looks large because price is above Rs. 6,000; MAPE comparable.
- ARIMA 30-day ranges nearly flat; HDFC Rs. 990.10 to 990.52 confirms its short-horizon limit.

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SHAP EXPLAINABILITY ANALYSIS



SHAP feature importance, mean |SHAP|

Feature-level attribution turns a model output into a decision-support explanation for a retail investor.

Rank	TVS Motor	Solar Ind.	HDFC Bank
1	MA200 (1.014)	MA200 (0.786)	MA200 (1.498)
2	BB_Width (0.539)	Price_vs_MA200 (0.557)	MA50 (0.521)
3	BB_Upper (0.416)	BB_Mid (0.396)	BB_Mid (0.435)

TVS BUY instance: bullish MA200 +0.884, Price_vs_MA200 +0.445, MACD_Hist +0.355; bearish BB_Width -0.662, BB_Upper -0.526, BB_Lower -0.478.

MA200

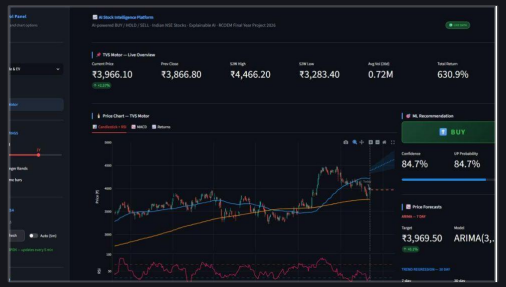
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DASHBOARD DEVELOPMENT

Streamlit + Plotly

Model output becomes a usable investor workflow.

From model output to decision support



Main dashboard view



Candlestick with ARIMA and 30-day trend forecast

Explore

Sector selector, time-period slider, Bollinger/volume toggles, manual and auto 5-minute refresh, market open/closed indicator.

Measure

Price, previous close, 52-week high/low, 20-day volume and total return.

Act with context

ML recommendation with confidence, RSI gauge, technical scoreboard and SHAP “Why this recommendation?” panel.

BACKTESTING RESULTS

The held-out test set shows where the strategy protects capital and where it misses persistent trends.

Stock	Start capital	Final value	Model return	Buy and hold	Trades	Win rate
TVS Motor	Rs. 1,00,000	Rs. 1,13,676	+13.7%	+14.9%	197	50.3%
Solar Industries	Rs. 1,00,000	Rs. 1,03,686	+3.7%	+34.2%	190	50.5%
HDFC Bank	Rs. 1,00,000	Rs. 91,601	-8.4%	-23.4%	71	40.8%

15 PTS

capital preserved by HDFC Bank versus buy-and-hold

Strategy rule

Buy when UP probability > 55%; hold cash when < 45%.

TVS Motor nearly matched buy-and-hold, only 1.2 points lower. Solar Industries lagged because cash periods missed a persistent bull run. Overall, the model is better at limiting losses in falling markets than at beating strong uptrends.

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DIFFICULTIES ENCOUNTERED AND RESOLUTIONS

The pipeline became reliable by resolving problems phase by phase, from data preparation to model delivery.

Phase	Problem	Resolution
3	Close column loaded as string from yfinance metadata rows	Custom loader with pd.to_numeric() and row filtering
3	Date index loaded as plain strings	Explicit pd.to_datetime() conversion
6	Accuracy 32-41% due to NaN rows from MA200	Strict dropna() before training
6	3-class accuracy 35-52%; HOLD class dominated	Switched to binary UP/DOWN; HOLD from confidence band
7	ARIMA: no supported index	asfreq("B") business-day frequency
8	pandas 2.x removed .last()	Replaced with .tail()
8	Empty data outside market hours	len(df) checks and clear error messages
8	ARIMA 30-day forecast too flat	Added Linear Regression trend model

01

DATA

Clean and index

Types, dates, missing values.

02

MODEL

Prevent leakage

Chronological split and balanced labels.

03

DELIVERY

Make it usable

Clear errors, robust forecasts and a dashboard.

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CLOSING REVIEW · PLATFORM ARC

WHAT THE PLATFORM DELIVERS

From market data to an accessible decision-support workflow.

Prepare

Data preparation across historical Indian market data.

Engineer

Feature engineering turns prices into technical indicators.

Compare

Model comparison across Gradient Boosting, ARIMA and LSTM.

Explain

SHAP makes the signals inspectable.

Present

Dashboard development turns outputs into a usable workflow.

Backtest

Backtesting connects model behaviour to decisions.

COMPLETED ARC · PREDICTION → INTERPRETATION → DECISION SUPPORT

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CONCLUSION

FROM MODELS TO DECISIONS

A deployable, explainable decision direction for Indian retail-market analysis

WHAT IS COMPLETE

Phases 2 to 8 are complete; Phase 9, Streamlit Cloud deployment, is in progress.

The platform combines

- Predictive models and model comparison
- SHAP explanations for interpretable signals
- A dashboard designed for practical usability
- Backtesting to examine decision behaviour

PERFORMANCE IS EVIDENCE · USABILITY IS THE DELIVERY TEST

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INTERPRETATION BOUNDARIES

LIMITATIONS THAT SHAPE INTERPRETATION

The results are useful within a defined historical and technical boundary.

1

No tick-by-tick data

The platform uses end-of-day market data, not tick-by-tick streaming; paid APIs would be needed for that layer.

2

No news or sentiment

News sentiment and fundamental data are outside the current technical-analysis scope.

3

No transaction-cost modelling

Backtesting does not model transaction costs; returns should not be read as realised investment outcomes.

4

Limited stock coverage

The current platform covers three stocks; findings do not represent the full Indian market.

5

Limited historical scope

The evidence comes from a limited historical window and cannot establish performance across every market regime.

6

Deployment still in progress

There is no live deployment yet. The LSTM model is not saved; the dashboard currently uses a linear estimate.

NEXT BUILD CYCLE

FUTURE SCOPE: THE NEXT BUILD CYCLE

A practical sequence from current prototype to broader decision support.

01

Deploy

Finish cloud deployment and move the dashboard toward live access.

02

Enrich

Add richer sentiment or news signals, including FinBERT news sentiment.

03

Broaden

Expand stock coverage beyond the current three-stock scope.

04

Improve

Continue improving the modelling and dashboard, including LSTM inference and P/E and earnings data.

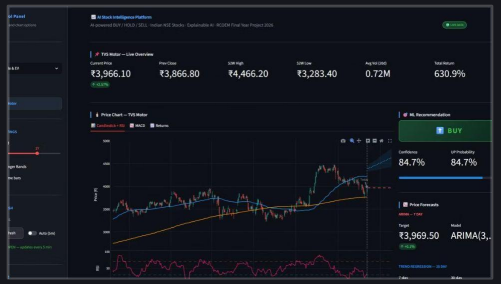
PHASE 9 IN PROGRESS · THE NEXT BUILD EXTENDS THE SAME EXPLAINABILITY PRINCIPLE

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OUTPUT SCREENSHOTS

OUTPUT SCREENSHOTS AND FINAL TAKEAWAY

The output area is shown through the dashboard views produced for the project. The screenshots remain evidence of the interface, not a substitute for the underlying analysis.



Main dashboard view



Candlestick with ARIMA and 30-day trend forecast

OUTPUT EVIDENCE · DASHBOARD + FORECAST VIEW

EXPLAINABLE AI-BASED SUPPORT PLATFORM

For Indian retail-market analysis.

TECHNICAL APPENDIX · FEATURE ENGINEERING

TECHNICAL INDICATORS: THE FORMULAS BEHIND THE SIGNALS

TREND

Simple moving average

$$MA(n)_t = \frac{1}{n} \times \sum_{i=0}^{n-1} Close_{t-i}$$

Close is the closing price; *n* is the lookback window. MA50 and MA200 smooth medium- and long-horizon movement.

VOLATILITY

Bollinger Bands

$$Middle = MA20$$

$$Upper = MA20 + 2\sigma_{20}$$

$$Lower = MA20 - 2\sigma_{20}$$

sigma is the standard deviation of price over the window. The bands expand and contract with volatility.

RETURN

Daily return

$$r_t = \frac{Close_t}{Close_{t-1}} - 1$$

*r*t compares today's close with the prior close. Rolling return statistics support performance and risk comparisons.



TECHNICAL APPENDIX · MODEL INPUTS

MOMENTUM, VOLATILITY AND MODEL INPUTS

01

MOMENTUM

RSI

$$\mathrm{RSI} = 100 - \frac{100}{1+\mathrm{RS}}$$

$$\mathrm{RS} = \frac{\text{average gain}}{\text{average loss}}$$

RS is the average gain / average loss over the chosen window.

02

MOMENTUM

MACD

$$\mathrm{MACD} = \mathrm{EMA12} - \mathrm{EMA26}$$

$$\mathrm{Signal} = \mathrm{EMA9}(\mathrm{MACD})$$

$$\mathrm{Histogram} = \mathrm{MACD} - \mathrm{Signal}$$

EMA is an exponential moving average.

03

DISPERSION

VOLATILITY

$$\sigma_n = \mathrm{std}(r_{t-n+1}, \ldots, r_t)$$

Volatility is the standard deviation of the last *n* returns. Together with volume and return features, these indicators feed the comparison models.

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CHALLENGES FACED ACROSS THE PROJECT

Each difficulty required a concrete control in the data, modeling or delivery layer.

Phase	Problem	Practical response			
DATA	Inconsistent data types, date indexes, missing values and empty datasets interrupted analysis.	Convert types explicitly, normalize the datetime index, apply missing-value handling, and check dataset length before analysis.			
FEATURES	Rolling indicators create early missing rows, while future information can leak into the present.	Drop invalid rows after feature creation and preserve chronological train/test splits to prevent look-ahead leakage.			
LABELS	Class imbalance made modest multi-class accuracy difficult to interpret without overstating the result.	Use the binary UP/DOWN framing for training and treat HOLD as a confidence-band recommendation.			

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CHALLENGES FACED (CONTINUED)

Model and delivery difficulties needed controls that keep forecasts and dashboard outputs readable.

Phase	Problem	Practical response
MODELS	ARIMA produced index warnings and short-horizon forecasts that were too flat.	Use a supported business-day frequency, constrain the short forecast horizon, and add a linear trend model for the longer view.
DELIVERY	Connecting trained models to a usable Streamlit + Plotly dashboard risked making metrics and recommendations hard to read.	Keep the dashboard layers explicit: model output, evaluation metrics and recommendations remain visible and understandable.

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TECHNICAL APPENDIX · NEXT STEPS

FUTURE OUTLOOK

Turning the analysis prototype into a dependable research tool for stock analysis.

1 Stronger models

Benchmark ensemble and deep-learning forecasts against the current ARIMA and linear baselines.

2 Live data pipeline

Automate daily market data ingestion, validation and retraining so signals stay current.

3 Evaluation and trust

Add backtesting with transaction costs, drift monitoring and clear confidence ranges on every signal.

4 User-ready dashboard

Extend the Streamlit + Plotly dashboard with alerts, portfolio views and exportable reports.

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TECHNICAL APPENDIX · CLOSING

THANK YOU

Questions and discussion welcome.

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